Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Let fx is continuous on $a,\quad c$ and differentiable in $a,\quad c$ where $a,\quad b,\quad c$ be real numbers; $a < b < c$. If $f^{'}x$ is strictly increasing function and c - $bfa + b$ - $afc = k$, then

(A) $k > (c-a)f(b)$

(B) $k < (c-a)f(b)$

(C) $k = (c-a)f(b)$

(D) $k \leq 2(c-a)f(b)$

Q.2

Let $f\left(x\right) = xx^2 + mx + n + 2,$ for all $x \in R$ and $m, n \in R$. If Rolle's Theorem holds for $f\left(x\right)$ at $x = \frac{4}{3}$ in $x \in [1,2],$ then $\left(m+n\right)$ equals

(A) 1

(B) 2

(C) 3

(D) 4

Q.3

If the function $\mathrm{f}:[0,16] \rightarrow \mathrm{R}$ is differentiable. If $0 < \alpha < 1$ and $1 < \beta < 2$, then $\int_0^{16} \mathrm{f}(\mathrm{t})\mathrm{dt}$ is equal to

(A) $4\left(\alpha^3 \mathrm{f}\left(\alpha^4\right) - \beta^3 \mathrm{f}\left(\beta^4\right)\right)$

(B) $4\left(\alpha^3 \mathrm{f}\left(\alpha^4\right) + \beta^3 \mathrm{f}\left(\beta^4\right)\right)$

(C) $4\left(\alpha^4 \mathrm{f}\left(\alpha^3\right) + \beta^4 \mathrm{f}\left(\beta^3\right)\right)$

(D) $4\left(\alpha^2 \mathrm{f}\left(\alpha^2\right) + \beta^2 \mathrm{f}\left(\beta^2\right)\right)$

Q.4

The value of c for which the conclusion of Lagrange's mean value theorem holds for the function $f(x) = \sqrt{25 - x^2}$ on the interval $[1,5]$ is

(A) $\sqrt{3}$

(B) $\sqrt{5}$

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(C) $\sqrt{15}$

(D) 2

Q.5

Statement-1: If $f(x) = e^{(x-1)(x-3)}$ then Rolle's theorem is applicable to $f(x)$ in the interval $[1, 3]$.
Statement-2: Mean value theorem is applicable to $f(x) = e^{(x-1)(x-3)}$ in the interval $[1, 4]$.

(A) STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Q.6

Let f be differentiable for all x. If $f(1) = -2$ and $f'(x) \geq 2$ for $x \in [1, 6]$ then

(A) $f(6) < 8$

(B) $f(6) \geq 8$

(C) $f(6) \geq 10$

(D) $f(6) \leq 5$

Q.7

Statement-1: $|\cot x - \cot y| \leq |x - y|$ for all $x, y \in (-\pi/2, \pi/2)$
Statement-2: If f is differentiable on an open interval and $|f'(x)| \leq M$ then $|f(x) - f(y)| \leq M\,|x - y|$

(A) STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Q.8

The absolute minimum value of $x^4 - x^2 - 2x + 5$

(A) is equal to 5

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(B) is equal to 3

(C) is equal to 7

(D) does not exist

Q.9

Let $f(x)$ and $g(x)$ be defined and differentiable for $x \geq x_0$ and $f(x_0) = g(x_0)$, $f'(x) > g'(x)$ for $x > x_0$ then

(A) $f(x) < g(x)$ for some $x > x_0$

(B) $f(x) = g(x)$ for some $x > x_0$

(C) $f(x) > g(x)$ for all $x > x_0$

(D) none of these

Q.10

If $f(x) = \log x$ satisfies Lagrange's theorem on $[1, e]$ then value of $c \in (1, e)$ such that the tangent at c is parallel to line joining $(1, f(1))$ and $(e, f(e))$ is

(A) $e - \frac{3}{2}$

(B) $\frac{1+e}{2}$

(C) $e - 1$

(D) $e - \frac{1}{2}$

Q.11

Let $g(x) = (\log(1+x))^{-1} - x^{-1}, x > 0$ then

(A) $1 < g(x) < 2$

(B) $-1 < g(x) < 0$

(C) $0 < g(x) < 1$

(D) none of these

Q.12

Let $f(x) = (x-4)(x-5)(x-6)(x-7)$ then

(A) $f'(x) = 0$ has four real roots

(B) three roots of $f'(x) = 0$ lie in $(4,5) \cup (5,6) \cup (6,7)$

(C) the equation $f'(x) =$ has only two roots

(D) three roots of $f'(x) = 0$ lie in $(3,4) \cup (4,5) \cup (5,6)$

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Q.13

Let $f(x) = \begin{cases} x^{\alpha} \log x & , \text{ if } x > 0 \\ 0 & , \text{ if } x = 0 \end{cases}$

If Rolle's theorem can be applied to f on $[0, 1]$ then value of α can be

(A) -1

(B) $-1/2$

(C) 0

(D) $1/2$

Q.14

The equation $e^{x-8} + 2x - 17 = 0$ has

(A) two real roots

(B) one real root

(C) eight real roots

(D) four real roots

Q.15

If Rolle's theorem holds for the function $f(x) = 2x^3 + ax^2 + bx$ in the interval $[-1, 1]$ for the point $c = \frac{1}{2}$, then the value of $2a + b$ is

(A) 1

(B) -1

(C) 2

(D) -2

Answers & Solutions

Q.1 Answer:

$k > c - afb$

Solution:

Using LMVT **on** fx **in** $a, \quad b$ **and** $b, \quad c$ **then**

$\frac{fb - fa}{b - a} = f'c_1, \quad c_1 \in a, \quad b$

$\frac{fc - fb}{c - b} = f'c_2, \quad c_2 \in b, \quad c$

$\therefore \quad a < c_1 < b < c_2 < c$

But $f'x \uparrow \Rightarrow f'c_1 < f'c_2$

$\frac{fb - fa}{b - a} < \frac{fc - fb}{c - b}$

$c - bfb - fa) < b - a(fc - fb$

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$\Rightarrow c - bfa + b - afc > c - afb$

Q.2 Answer:

3

Solution:

$f(1) = f(2)$

$\Rightarrow 3 \quad m + n + 7 = 0 \qquad \ldots\ldots (1)$

and $f'(x)$ **(at** $x = 4 \quad / \quad 3) = 0$ **gives**

$8 \quad m + 3 \quad n + 16 = 0 \qquad \ldots\ldots\ldots (2)$

$\Rightarrow m = -5, \quad n = 8$

Q.3 Answer:

$4\left(\alpha^3 f\left(\alpha^4\right) + \beta^3 f\left(\beta^4\right)\right)$

Solution:

$I = \int_0^{16} f(t)dt$

Consider $g(x) = \int_0^{x^4} f(t)dt \quad \Rightarrow \quad g(0) = 0$

LMVT for g in $[0, 1]$ **gives, some** $\alpha \in (0, 1)$ **such that**

$\frac{g(1)-g(0)}{1-0} = g'(\alpha)$ **...(1)**

LMVT in $[1, 2]$ **gives some** $\beta \in (1, 2)$ **such that**

$\frac{g(2)-g(1)}{2-1} = g'(\beta)$ **...(2)**

(1) + (2)

$g'(\alpha) + g'(\beta) = g(2) - \underbrace{g(0)}_{\text{zero}};$

But $g'(x) = f\left(x^4\right) \cdot 4x^3$

$4\left(\alpha^3 f\left(\alpha^4\right) + \beta^3 f\left(\beta^4\right)\right) = \int_0^{16} f(t)dt$

Q.4 Answer:

$\sqrt{15}$

Solution:

$f(5) = 0, f(1) = \sqrt{24} \cdot \frac{f(5)-f(1)}{5-1} = \frac{1}{4}\sqrt{24} = \sqrt{\frac{3}{2}}$

$f'(x) = \frac{-x}{\sqrt{25-x^2}}$ **so** $f'(c) = \sqrt{\frac{3}{2}} \Rightarrow \frac{c^2}{25-c^2} = \frac{3}{2}$

$\Rightarrow 2c^2 = 75 - 3c^2 \Rightarrow c^2 = 15.$

For $c \in (1, 5)$, **we have** $c = \sqrt{15}.$

Q.5 Answer:

STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

Solution:

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f in statement- 1 is continuous function on $[1,3]$ and differentiable on $(1,3) f(3)=1=f(1)$ so Rolle's theorem is applicable.

Again the mean value is applicable so statement- 2 is true but is not a correct explanation for statement-1.

Q.6 Answer:

$f(6) \geq 8$

Solution:

Applying Lagrange's mean value theorem there is $c \in (1,6)$
such that $\frac{f(6)-f(1)}{5}=f'(c) \geq 2$
$\Rightarrow f(6) \geq f(1)+10=8.$

Q.7 Answer:

STATEMENT-1 is False, STATEMENT-2 is True

Solution:

Applying Lagrange's mean value on $[x, y]$, we have $c \in (x, y)$ such that
$f'(c)=\frac{f(y)-f(x)}{y-x}$
$|f(y)-f(x)|=|f'(c)|\,|y-x| \leq M|y-x|$
(if \mid $f'(x)\,| \leq M$)
and $|f'(y)-f(x)| \geq M|x-y|$ (if $|f'(x)| \geq M$)
Putting $f(x)=\cot x, f'(x)=-\operatorname{cosec}^2 x,|f'(x)| \geq 1$
Hence $|\cot x-\cot y| \geq |x-y|$

Q.8 Answer:

is equal to 3

Solution:

$$f(x)=x^4-x^2-2x+5$$
$$f'(x)=4x^3-2x-2$$
$$=(x-1)\left(4x^2+4x+2\right)$$

Clearly at $x=1$, we will set the minimum value which is 3.

Q.9 Answer:

$f(x)>g(x)$ for all $x>x_0$

Solution:

Consider the function $\phi(x)=f(x)-g(x)$ on the interval $[x_0, x]$. By Lagrange's theorem we have $\phi(x)-\phi\left(x_0\right)=\phi'(z)\left(x-x_0\right)$ for some $z \in (x_0, x)$.
Since $\phi\left(x_0\right)$

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$= 0, \phi'(z) = f'(z) - g'(z) > 0.$
So $\phi(x) - \phi(x_0) = \phi(x)$
$= (f'(z) - g'(z))(x - x_0) > 0 \Rightarrow \phi(x) > 0$
Thus $f(x) > g(x)$ **for all** $x > x_0$**.**

Q.10 Answer:

$e - 1$

Solution:

$f'(c) = \frac{1}{c}$
So $\frac{f(e)-f(1)}{e-1} = f'(c) = \frac{1}{c}$
$\Rightarrow \quad \frac{\log e - \log 1}{e-1} = \frac{1}{c}$
$\Rightarrow \quad \frac{1}{e-1} = \frac{1}{c} \quad \Rightarrow \quad c = e - 1$

Q.11 Answer:

$0 < g(x) < 1$

Solution:

For $x > 0$**, consider the function** $f(x) = \log(1 + x)$ **on the interval** $[0, x]$**.**
Since the function is differentiable on $(0, x)$**, by Lagrange's theorem, there is a** $\xi \in (0, x)$ **such that**

$$\frac{\log(1+x) - \log(1+0)}{x - 0} = f'(\xi) = \frac{1}{1+\xi}$$

$$\Rightarrow \quad \frac{\log(1+x)}{x} = \frac{1}{1+\xi} < 1$$

$$\Rightarrow \quad \log(1+x) < x \quad \Rightarrow \quad g(x) > 0$$

Also $\xi \in (0, x) \Rightarrow \xi < x \Rightarrow \frac{1}{1+\xi} > \frac{1}{1+x}$
$\Rightarrow \quad \frac{\log(1+x)}{x} > \frac{1}{1+x}$
$\Rightarrow \quad \log(1+x) > \frac{x}{1+x} \Rightarrow g(x) < 1.$

Q.12 Answer:

three roots of $f'(x) = 0$ lie in $(4, 5) \cup (5, 6) \cup (6, 7)$

Solution:

Since $f(4) = f(5) = f(6) = f(7) = 0$**, so by Rolle's theorem applied to the intervals** $[4, 5], [5, 6], [6, 7]$ **there exist** $x_1 \in (4, 5), x_2 \in (5, 6), x_3 \in (6, 7)$ **such that** $f'(x_1) = f'(x_2) = f'(x_3) = 0$**. Since** f' **is a polynomial of degree 3 so cannot have four roots.**

Q.13 Answer:

$1/2$

Solution:

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For the Rolle's theorem to be applicable on $[0,1]$, we must have that f is continuous on $[0,1]$ and in particular at $x=0$ from the right. So $\lim_{x\to 0+} x^{\alpha}\log x = f(0) = 0$

$\Rightarrow \quad \lim_{x\to 0+} \frac{\log x}{x^{-\alpha}} = 0$

$\Rightarrow \quad \lim_{x\to 0+} \frac{(1/x)}{(-\alpha)x^{-\alpha-1}} = 0 \Rightarrow \lim_{x\to 0+} \frac{x^{\alpha}}{-\alpha} = 0$

This is possible only if $\alpha > 0$.

Q.14 Answer:

one real root

Solution:

Clearly $x=8$ satisfies the given equation.

Assume that $f(x) = e^{x-8} + 2x - 17 = 0$ has a real root α other than $x=8$.

We may suppose that $\alpha > 8$ (the case for $\alpha < 8$ is exactly similar).

Applying Rolle's theorem on $[8,\alpha]$, we get $\beta \in (8,\alpha)$, such that $f'(\beta)=0$.

But $f'(\beta) = e^{\beta-8} + 2$, so that $e^{\beta-8} = -2$ which is not possible.

Hence there is no real root other than 8.

Q.15 Answer:

-1

Solution:

$f(-1) = f(1)$

$\Rightarrow -2 + a - b = 2 + a + b$

$\Rightarrow b = -2$

Also, $f'(c) = 0 \Rightarrow 6c^2 + 2ac + b = 0$

$\Rightarrow \frac{3}{2} + a - 2 = 0 \Rightarrow a = \frac{1}{2}$

Thus, $2a + b = 1 - 2 = -1$

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